

Wind-Assisted Propulsion Integration

Feasibility, Structural Integration, and Performance Modeling for Bulker Fleet Retrofits

Author: Andrei Simakin

Organization: Fleet Experts Solutions

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1. Overview of Wind-Assisted Propulsion (WAPS)

As commercial shipping transitions toward zero-emission power, Wind-Assisted Propulsion Systems (WAPS) — including Flettner Rotor Sails, Rigid Wings, and Suction Sails — provide immediate, renewable thrust directly off the wind. On favorable trade routes (such as North Atlantic, Trans-Pacific, and Australia–Asia bulk trade lanes), WAPS delivers substantial fuel and emissions savings.

8% – 20%

NET ANNUAL FUEL SAVINGS

2 - 4 Units

TYPICAL BULKER SETUP

Class Approved

DNV / LR / BV / ABS

2. Key Engineering Considerations

2.1 Structural & Deck Integration

Installation of Rotor Sails requires local reinforcement of main deck structures, hatch coamings, and transverse bulkheads to withstand extreme Magnus force overturning moments during heavy weather. Foundation pedestals must be structurally tied to primary transverse deck strip plating.

2.2 Stability & Blind Sector Navigation

- **Intact Stability:** Comprehensive heel angle calculations under maximum aerodynamic windage area.
- **Bridge Navigational Visibility:** Compliance with SOLAS Chapter V Regulation 22 regarding forward view obstruction and radar blind arcs.
- **Cargo Operations:** Tilting or foldable foundation mechanisms required for unhindered crane and shore loader clearance during port loading/unloading.

3. Technology Comparison

Technology Type	Operating Principle	Typical Lift Coeff (C_L)	Operational Footprint
Rotor Sails (Flettner)	Magnus effect on rotating cylinder	10.0 – 12.5	Compact circular footprint
Rigid Wings (Suction)	Boundary layer suction aerofoil	5.0 – 7.0	Folding vertical wing structure
Solid Wingsails	Traditional aerofoil element	2.0 – 3.5	Requires active reefing/feathering

Technical Service Note: Fleet Experts Solutions prepares full 3D FEM structural strength analyses, stability booklet revisions, and Class approval packages for WAPS integration.

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